

Statistical modeling

Module 06

Marc-Olivier Boldi

Bachelor, GSEM

Autumn 2018

Inference on each coefficient

In the research context, we have seen ANOVA table, F-tests, and model comparisons. These methods allow to test whether a variable is significant in a model.

Research question can also be related to testing some coefficient individually, especially when significance of the difference in levels of categorical variables are of interest.

To do this, we use the t-tests for the coefficients of the regression.

Inference on each coefficient

Example: Whitening power of 4 different toothpastes.

```
> fit <- lm(Whiteness~Toothpaste, data=toothpaste)
> summary(fit)
```

Call:

```
lm(formula = Whiteness ~ Toothpaste, data = toothpaste)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-9.1667	-1.6250	-0.0833	1.6250	5.8333

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.000	1.419	13.389	1.92e-11 ***
ToothpasteT2	1.667	2.007	0.830	0.416085
ToothpasteT3	9.167	2.007	4.568	0.000187 ***
ToothpasteT4	5.500	2.007	2.741	0.012607 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
[...]

Inference on each coefficient

The regression equation is

$$y = b_0 + b_{T2}\delta_{T2}(x) + b_{T3}\delta_{T3}(x) + b_{T4}\delta_{T4}(x) + \epsilon.$$

The estimates can be interpreted as:

- ▶ Average y
 - ▶ for T1 is 19,
 - ▶ for T2 is 20.67,
 - ▶ for T3 is 28.17,
 - ▶ for T4 is 24.5.
- ▶ Average difference between
 - ▶ T2 and T1 is 1.67,
 - ▶ T3 and T1 is 9.17,
 - ▶ T4 and T1 is 5.5.

Inference on each coefficient

The numbers below column $\Pr(>|t|)$ are the p -values of the t-tests of each coefficients. Each one tests if the coefficient is different from 0:

If the p -value is greater than 0.05, the coefficient is declared non-significantly different from zero.

Notes:

- ▶ The significance level 0.05 is ad-hoc. It is the most used value and, except mentioned otherwise, the default in this course.
- ▶ For a quick reading, stars indicate the significance level.
- ▶ A classical language shortcut is to say "the variable is not-significant". But it should be clearly understood that it refers to the coefficient comparison to 0.

Inference on each coefficient

Example: b_{T2} is not significantly different from 0 (at 5%). The others are.

Conclusions: there is no significant difference between the whitening power of T1 and T2. There is a significant difference between T1 and T3, and between T1 and T4.

This kind of conclusions could not have been obtained from an F-test.

On which theory is this test built?

Distribution of the estimates

Suppose that, in the population, the regression is

$$y = b_0 + b_1x_1 + \cdots + b_px_p + \epsilon.$$

In the sample, the (estimated) regression is

$$y = \hat{b}_0 + \hat{b}_1x_1 + \cdots + \hat{b}_px_p + \epsilon.$$

Then it can be shown¹ that ,

$$\hat{b}_j \sim N(b_j, v_{jj}\sigma^2) \quad \text{for } j = 0, \dots, p.$$

where

- ▶ v_{jj} is $(X^T X)^{-1}_{jj}$ the j -th element of the diagonal of $(X^T X)^{-1}$,
- ▶ $v_{jj'}$ is $(X^T X)^{-1}_{jj'}$ the (j, j') -th element of $(X^T X)^{-1}$.

¹Under the model assumptions.

Distribution of the estimates and its estimates...

The estimate of the standard deviation of $\hat{b}_j$, $\sqrt{v_{jj}}\sigma$, is

$$s_{\hat{b}_j} = \sqrt{v_{jj}}s$$

It is called the standard error. It appears below "Std. Error".

The t -value is defined as

$$t_j = \frac{\hat{b}_j}{s\sqrt{v_{jj}}}.$$

It appears below "t value". See the example.

Individual coefficient tests

The null hypothesis of the test H_0 is

$$H_0 : b_j = 0$$

Under H_0 , it can be shown that

$$t_j \sim t_{n-p-1}.$$

A statistical test at level α for

$$H_0 : b_j = 0 \quad \text{versus} \quad H_1 : b_j \neq 0$$

can be built by rejecting H_0 if (equivalent):

- ▶ $|t_j| > t_{n-p-1; 1-\alpha/2},$
- ▶ $p_j = 2 - 2 \times t_{n-p-1}(t_j) < \alpha.$

See column $\Pr(>|t|)$ in the example.

Individual coefficient tests

Example: Another example, the burn out case.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	291.70648	143.20669	2.037	0.0610 .
Social	-4.61486	6.05980	-0.762	0.4590
I(Social^2)	0.10154	0.05730	1.772	0.0981 .
SexM	-142.11265	267.11187	-0.532	0.6030
Social:SexM	0.21461	11.15342	0.019	0.9849
I(Social^2):SexM	0.05772	0.10136	0.569	0.5781

Conclusion: except the slope of Social, at 5%, any coefficient can be set to 0 **keeping all the others.**

Individual coefficient tests

Important remark: the t-test are individuals.

One of the biggest possible mistakes in this course² is to declare **all** the variables not-significant.

The t-test allows to test the significance of each variable **keeping all the others**. In particular, it is possible that among two variables appearing as non-significant with the t-test, at least one would be significant if the other was removed from the model.

For example, it **could be** that `Social` becomes significant if $I(\text{Social}^2)$ was removed from the model. To know this, one needs to try. See later.

²and in the exam...

Individual coefficient tests vs model comparisons

Individual tests cannot be used to test several coefficients at once. For this, use model comparisons or ANOVA instead.

For example, to test if all the coefficients associated to Sex are not significant, compare the models with and without it:

```
> fit1 <- lm(BurnOut~(Social+I(Social^2))*Sex, data=ashley)
> fit0 <- lm(BurnOut~Social+I(Social^2), data=ashley)
> anova(fit0, fit1)
Analysis of Variance Table
```

```
Model 1: BurnOut ~ Social + I(Social^2)
Model 2: BurnOut ~ (Social + I(Social^2)) * Sex
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	17	361115				
2	14	127317	3	233797	8.5696	0.001769 **

Conclusion: we cannot remove **all** the coefficients associated with factor Sex.

Confidence interval

Another important application of the limit distribution is the construction of confidence intervals.

A confidence interval at level $1 - \alpha$ for b_j can be built from the limit distribution of the estimate $\hat{b}_j$:

$$\frac{\hat{b}_j - b_j}{\sigma \sqrt{v_{jj}}} \sim N(0, 1)$$

Therefore, with z_α the α -quantile of $N(0, 1)$,

$$\begin{aligned} 1 - \alpha &= P \left(z_{\alpha/2} \leq \frac{\hat{b}_j - b_j}{\sigma \sqrt{v_{jj}}} \leq z_{1-\alpha/2} \right) \\ &= P \left(\hat{b}_j + z_{\alpha/2} \sigma \sqrt{v_{jj}} \leq b_j \leq \hat{b}_j + z_{1-\alpha/2} \sigma \sqrt{v_{jj}} \right) \end{aligned}$$

Normal confidence interval at $1 - \alpha$:

$$\left[\hat{b}_j + z_{\alpha/2} s \sqrt{v_{jj}}, \hat{b}_j + z_{1-\alpha/2} s \sqrt{v_{jj}} \right].$$

t -confidence interval

Looking carefully at the previous development, the distribution uses σ , while the CI uses its estimate s , since σ is not known.

This has consequences and, in fact, the normal CI is only approximate. A better CI can be built. Indeed, It can be shown that

$$\frac{\hat{b}_j - b_j}{s\sqrt{v_{jj}}} \sim t_{n-p-1}.$$

Thus a confidence interval at $1 - \alpha$ can be built:

$$\left[\hat{b}_j + t_{n-p-1;\alpha/2} s\sqrt{v_{jj}}, \hat{b}_j + t_{n-p-1;1-\alpha/2} s\sqrt{v_{jj}} \right].$$

The confidence interval is interpreted as a range of precision on the estimate of the coefficient with a given confidence level.

t-confidence interval: examples

Toothpaste:

```
> confint(fit)
                2.5 %      97.5 %
(Intercept)  16.039777 21.960223
ToothpasteT2 -2.519721  5.853054
ToothpasteT3  4.980279 13.353054
ToothpasteT4  1.313612  9.686388
```

Burn out:

```
> confint(fit1)
                2.5 %      97.5 %
(Intercept)    -15.4413154 598.8542795
Social          -17.6118401   8.3821200
I(Social^2)      -0.0213515   0.2244359
SexM            -715.0106328 430.7853391
Social:SexM      -23.7070875  24.1363102
I(Social^2):SexM -0.1596726   0.2751168
```

Confidence intervals

Which one to use?

- ▶ The t interval is more accurate.
- ▶ For large n compare to p , the two intervals match quite closely and which one to use is not so important in practice.
- ▶ With a statistical program (R, SPSS, etc.) the t interval is given in general.
- ▶ In practice they often give very close results and the normal one is easier to use (e.g. with a calculator, Excel, etc.).

In any case, the confidence intervals should not be confounded with prediction intervals:

- ▶ Confidence intervals are on the model coefficients (b).
- ▶ Prediction intervals are on the response (y).

Tests of linear combinations of coefficients

Example: For the toothpaste

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	19.000	1.419	13.389	1.92e-11	***
ToothpasteT2	1.667	2.007	0.830	0.416085	
ToothpasteT3	9.167	2.007	4.568	0.000187	***
ToothpasteT4	5.500	2.007	2.741	0.012607	*

One can conclude that T2 is different (at 5%) of T1 (the reference).

How can we compare T2 and T3, for example?

Tests of linear combinations of coefficients

Translate this into statistical hypothesis. First note that:

$$T2 = b_0 + b_{T2}$$

$$T3 = b_0 + b_{T3}$$

Thus, the hypothesis comparing T2 and T3 can be rewritten as

$$T2 = T3 \iff T2 - T3 = 0 \iff b_{T2} - b_{T3} = 0$$

Therefore, we need to be able to test the linear combination $b_{T2} - b_{T3}$ using its estimate $\hat{b}_{T2} - \hat{b}_{T3}$.

Tests of linear combinations of coefficients

Let's look at the distribution of $\hat{b}_{T2} - \hat{b}_{T3}$:

$$\hat{b}_{T2} - \hat{b}_{T3} \sim N(b_{T2} - b_{T3}, \sigma_{T2-T3}^2)$$

where the estimate of σ_{T2-T3}^2 is

$$s_{T2-T3}^2 = s_{\hat{b}_{T2}}^2 + s_{\hat{b}_{T3}}^2 - 2s_{\hat{b}_{T2}}s_{\hat{b}_{T3}}\text{cor}(\hat{b}_{T2}, \hat{b}_{T3}).$$

Under the null hypothesis $H_0 : b_{T2} - b_{T3} = 0$, we have

$$\frac{\hat{b}_{T2} - \hat{b}_{T3}}{s_{T2-T3}} \sim t_{n-p-1}$$

Estimates of the correlations of the estimates...

We see in the previous formula that we need to be able to estimate $cor(\hat{b}_{T2}, \hat{b}_{T3})$, that is the correlation of the estimates.

The covariances and correlations between the coefficient estimates are

$$Cov(\hat{b}_j, \hat{b}_{j'}) = v_{jj'}\sigma^2, \quad Cor(\hat{b}_j, \hat{b}_{j'}) = \frac{v_{jj'}}{\sqrt{v_{jj}v_{j'j'}}$$

The estimates of the covariances and correlations between the coef. estimates are:

$$\hat{Cov}(\hat{b}_j, \hat{b}_{j'}) = v_{jj'}s^2, \quad \hat{Cor}(\hat{b}_j, \hat{b}_{j'}) = \frac{v_{jj'}}{\sqrt{v_{jj}v_{j'j'}}$$

Estimates of the correlations of the estimates...

These values appears in the summary of the model in R. Burn out example:

```
> summary(fit1, corr=TRUE)
[...]
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	291.70648	143.20669	2.037	0.0610 .
Social	-4.61486	6.05980	-0.762	0.4590
I(Social^2)	0.10154	0.05730	1.772	0.0981 .
SexM	-142.11265	267.11187	-0.532	0.6030
Social:SexM	0.21461	11.15342	0.019	0.9849
I(Social^2):SexM	0.05772	0.10136	0.569	0.5781

```
[...]
```

Correlation of Coefficients:

	(Intercept)	Social	I(Social^2)	SexM	Social:SexM
Social	-0.89				
I(Social^2)	0.78	-0.98			
SexM	-0.54	0.47	-0.42		
Social:SexM	0.48	-0.54	0.53	-0.93	
I(Social^2):SexM	-0.44	0.55	-0.57	0.86	-0.98

Tests of linear combinations of coefficients: example

Function `linearHypothesis` allows to extract all these elements and makes the test.

```
> linearHypothesis(fit, hypothesis.matrix="ToothpasteT2 - ToothpasteT3")
Linear hypothesis test
```

Hypothesis:

ToothpasteT2 - ToothpasteT3 = 0

Model 1: restricted model

Model 2: Whiteness ~ Toothpaste

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	410.42				
2	20	241.67	1	168.75	13.966	0.0013 **

Conclusion: the two toothpastes T2 and T3 are significantly different at 5%.

Technical details:

- ▶ The above test is a Wald test (somehow equivalent to the *t*-test)
- ▶ Other functions in R exist (`lht`, `glht`, etc.).

Tests of linear combinations of coefficients: general case

To go further...

Any linear combination of the coefficients can be tested:

$$a_0b_0 + \dots + a_pb_p.$$

Null hypothesis:

$$H_0 : a^T b = 0$$

where $a = (a_0, \dots, a_p)$. The statistic is:

$$\frac{a^T \hat{b}}{s \sqrt{a^T (X^T X)^{-1} a}} \sim t_{n-p-1}$$

Tests of linear combinations of coefficients: general case

Example: another way to test " $T2 - T3 = 0$ "

```
> linearHypothesis(fit, hypothesis.matrix=c(0,1,-1,0))
Linear hypothesis test
```

Hypothesis:

ToothpasteT2 - ToothpasteT3 = 0

Model 1: restricted model

Model 2: Whiteness ~ Toothpaste

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	410.42				
2	20	241.67	1	168.75	13.966	0.0013 **

Example: Test³ if b_{T3} is twice as large as b_{T4} .

```
> linearHypothesis(fit, hypothesis.matrix=c(0,0,1,-2))
Linear hypothesis test
```

Hypothesis:

ToothpasteT3 - 2 ToothpasteT4 = 0

Model 1: restricted model

Model 2: Whiteness ~ Toothpaste

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	21	245.03				
2	20	241.67	1	3.3611	0.2782	0.6037

³Indeed, it is useless; just here for illustration.